

STEFANO MICHELE HIDALGO - SERGIO VITTORIO ZAMBELLI - GIOVANNI D'AMBROSIO - FABIO CHIARINI

VALVE

2024

CORPORATION

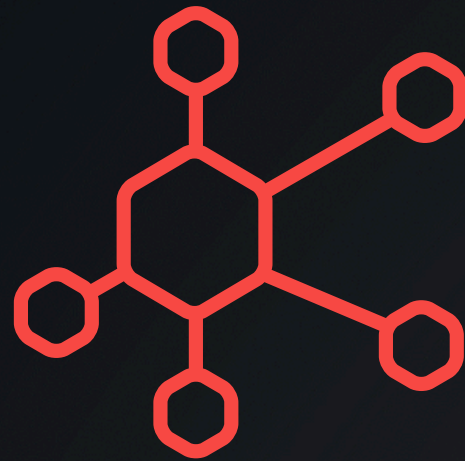
A FEARLESS ADVENTURE IN KNOWING WHAT TO DO WHEN NO ONE'S THERE TELLING YOU WHAT TO DO

.....

TABLE OF CONTENTS

01

CYSEC's Structure



03

SWOT Analysis



02

CYSEC's Culture



04

Planning



CORE BUSINESS OPERATIONS

**SOFTWARE
DEVELOPMENT
&
PUBLISHING**



**DIGITAL
DISTRIBUTION**

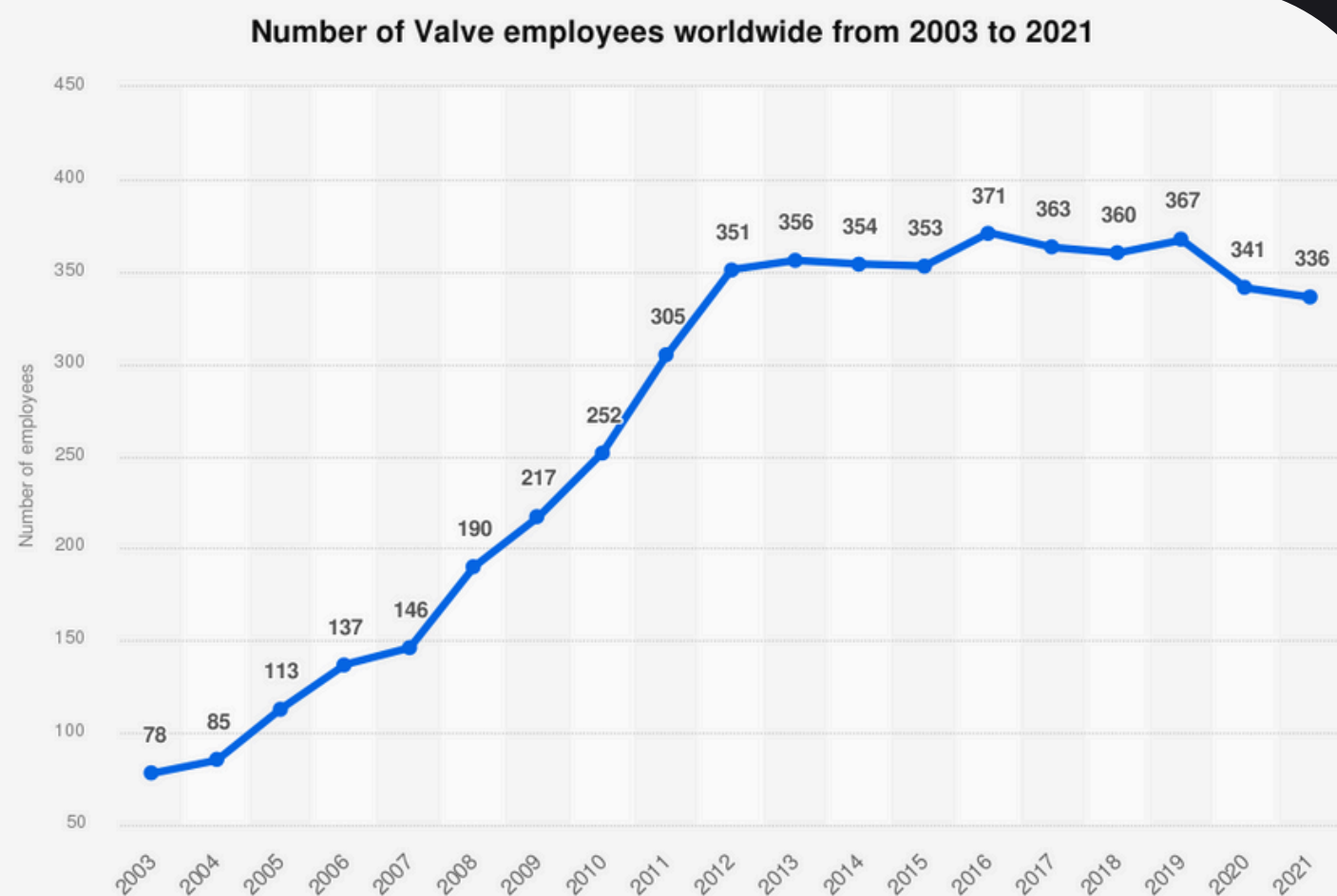


LARGE ENTERPRISE



VALVE CORPORATION - OFFICE LOCATIONS

- 📍 HQ - 10400 NE 4th St, Bellevue, WA 98004, US
- 📍 Rödingsmarkt 9, 20457 Hamburg, DE



Sources
Valve; The Verge
© Statista 2024

Additional Information:
Worldwide; Valve; 2003 to 2021



ESTIMATED REVENUE AT \$6.5B ^[2]

VALVE'S ORGANIZATIONAL STRUCTURE^{[3][4]}



FLAT ORGANIZATION WITH
NO MANAGEMENT
HIERARCHY



COMPLETE EMPLOYEE
FREEDOM IN PROJECT
SELECTION



"**RULE OF THREE**":
CONSENSUS OF AT LEAST 3
PEOPLE NEEDED TO START/
PUBLISH PROJECTS



EVALUATION SYSTEM BASED
ON PEER REVIEW AND
STACK RANKING

VALVE AS A POLYARCHY^{[3][4]}



Distinctive Features

- Social Entrepreneurship approach
- Complete team autonomy
- Organic formation of working groups ("Cabals")

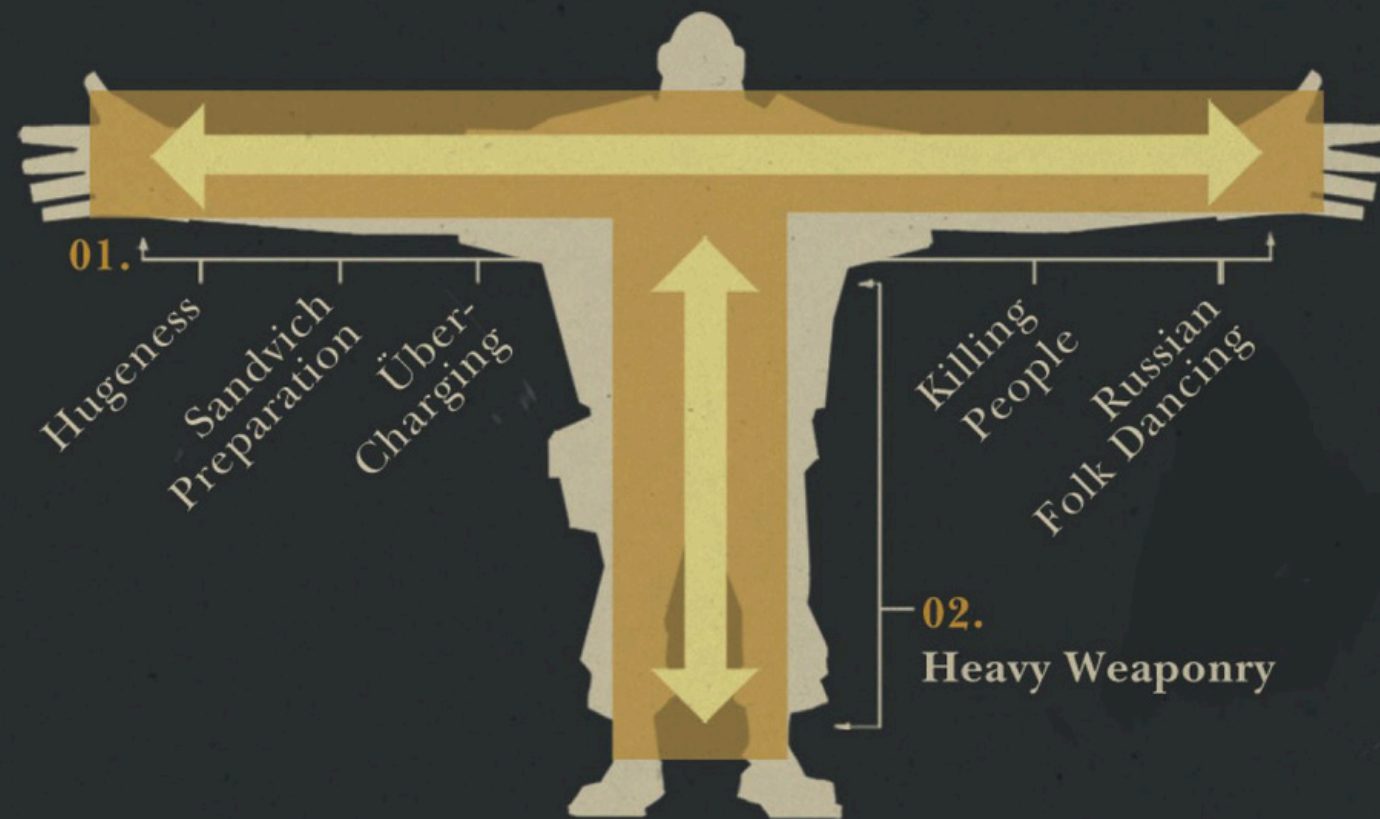


Impact on Cybersecurity

- No formal CISO or IT department
- Teams ("Cabals") form organically to handle incidents
- Collaborative security approach
- Distributed responsibility

HIRING AND TEAMS^{[3][4]}

T-SHAPED MODEL: EMPLOYEE



01. Broad-range generalist 02. Deep expertise in one area

"With the bar this high, would I be hired today?"

Organically formed teams, called Cabals

Valve's Handbook - Preface (2012)



CYBERSECURITY CULTURE

ENVIRONMENT
AS A RESOURCE



The environment should be
treated as a resource, not a
threat

SHARED
EXPERTISE



Having the support of users
allows leveraging the
knowledge of everyone

STRONG
AFFECTION



Impressive 75% of market
share in digital distribution
of PC games

ARTIFACTS

Support Site ^[5]

A platform dedicated to solving your cybersecurity problems



HackerOne ^[5]

A platform to solve bugs with external help



Response to ^{[6][7]}
community feedback

Hall of Fame ^[5]

Culture of collaboration and rewards those who assist



RESPONSIVENESS ^[8]

How the organization faced challenges



Hello x,

We are contacting you because you recently launched [REDACTED] on Steam. The Steam account for the developer of this game was recently compromised and the attackers uploaded a new build that contained malware.

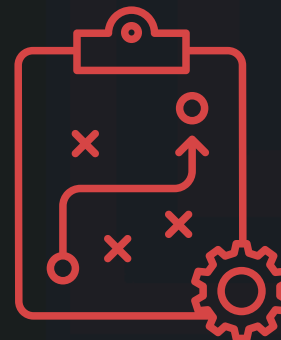
The compromised build was uploaded on Aug 24, 2023 @ 9:33AM PST and reverted on Aug 25, 2023 @ 12:06PM PST. If you played [REDACTED] on Steam during this time period, it is likely that malware infected your computer.

The build containing the suspected malware was promptly reverted and purged from Steam, but we strongly encourage you to run a full-system scan using an anti-virus product that you trust or use regularly, and inspect your system for unexpected or newly installed software. You may also consider fully reformatting your operating system to ensure that no malicious software remains on your machine.



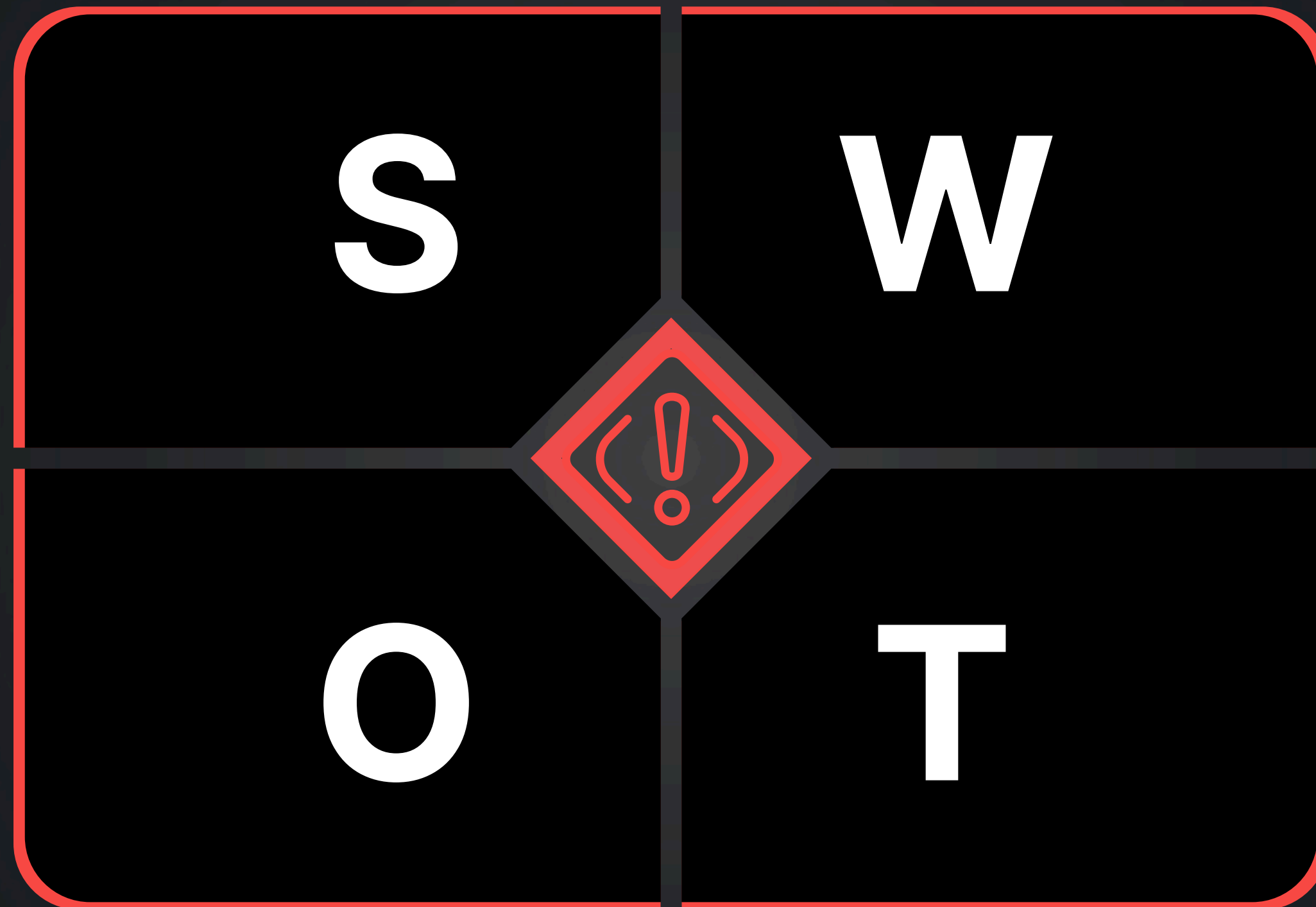
August 2023

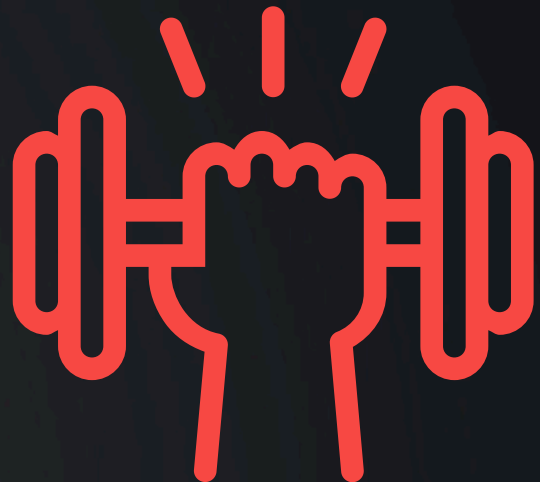
Attackers hacked developer accounts to spread malware through game updates on Steam



October 2023

Introduction of mandatory SMS-based 2FA for all developers attempting to publish updates or add new users to their accounts





S

Strengths ^[9]

- Strong bond and engagement from the community
- Big focus on Digital Rights Management and anti-cheat measures
- Highly specialized and talented employee base

Weaknesses ^[10]

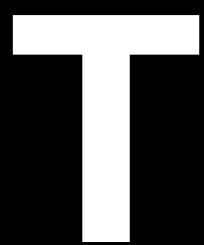
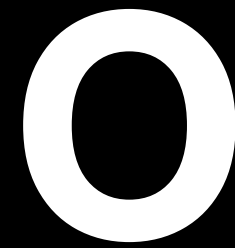
- Decentralized control could slow security protocol implementation during a cyber incident
 - High target for hackers
- Potential user compliance issue

W



Opportunities^[11]

- Implementation of AI-powered security technologies
- Employee-driven security standards
- User education programs



Threats^[10]

- Growing sophistication of threats
- Continuous exposure to phishing attacks
- Third-party vulnerabilities



“

In 1996, we set out to make great games, but we knew back then that we had to first create a place that was designed to foster that greatness. A place where incredibly talented individuals are empowered to put their best work into the hands of millions of people, with very little in their way.

”

Valve's Handbook - Preface (2012)

REACTIVE APPROACH

"WE DEVELOP CUTTING-EDGE GAMES AND
CROSS-PLATFORM GAME ENGINE
TECHNOLOGY!"

- Information Security Team
- Strategic approach
- Outsourcing for cybersecurity
- User awareness

What Is *Valve Not Good At?*

The design of the company has some downsides. We usually think they're worth the cost, but it's worth noting that there are a number of things we wish we were better at:

[4][10]



EPIC GAMES^[12]

"WE DEVELOP CUTTING-EDGE GAMES AND
CROSS-PLATFORM GAME ENGINE
TECHNOLOGY!"

- Information Security Team
- Strategic approach
- Outsourcing for cybersecurity
(partnership with RSI Security)
- User awareness

SECURITY DEPARTMENT

Chief Information
Security Officer
responsible for
protecting an
organization's data,
systems, and security
strategy.



OUTSOURCING

OUR
APPROACH

SECURITY DEPARTMENT

**Create a hierarchical
structure in order to
enhance cybersecurity**

SECURITY
DEPARTMENT

OUTSOURCING

Task of cybersecurity
assigned to an
external organization,
in order to pay full
attention to what
Valve can do best.



OUR
APPROACH

OUTSOURCING

**Knock at the door of
another company for
them to solve your
problem**

SECURITY
DEPARTMENT

OUTSOURCING

CENTER OF EXCELLENCE

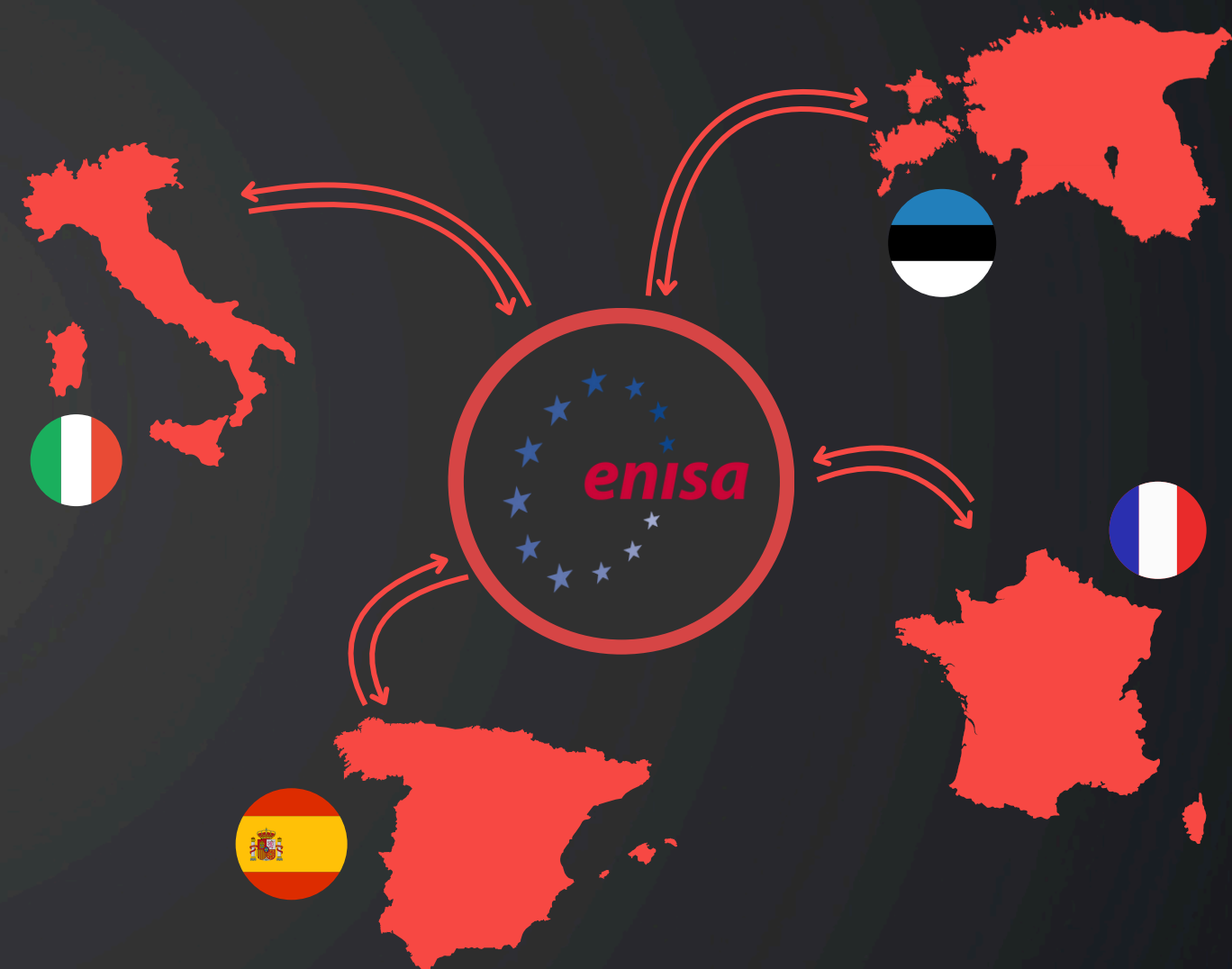
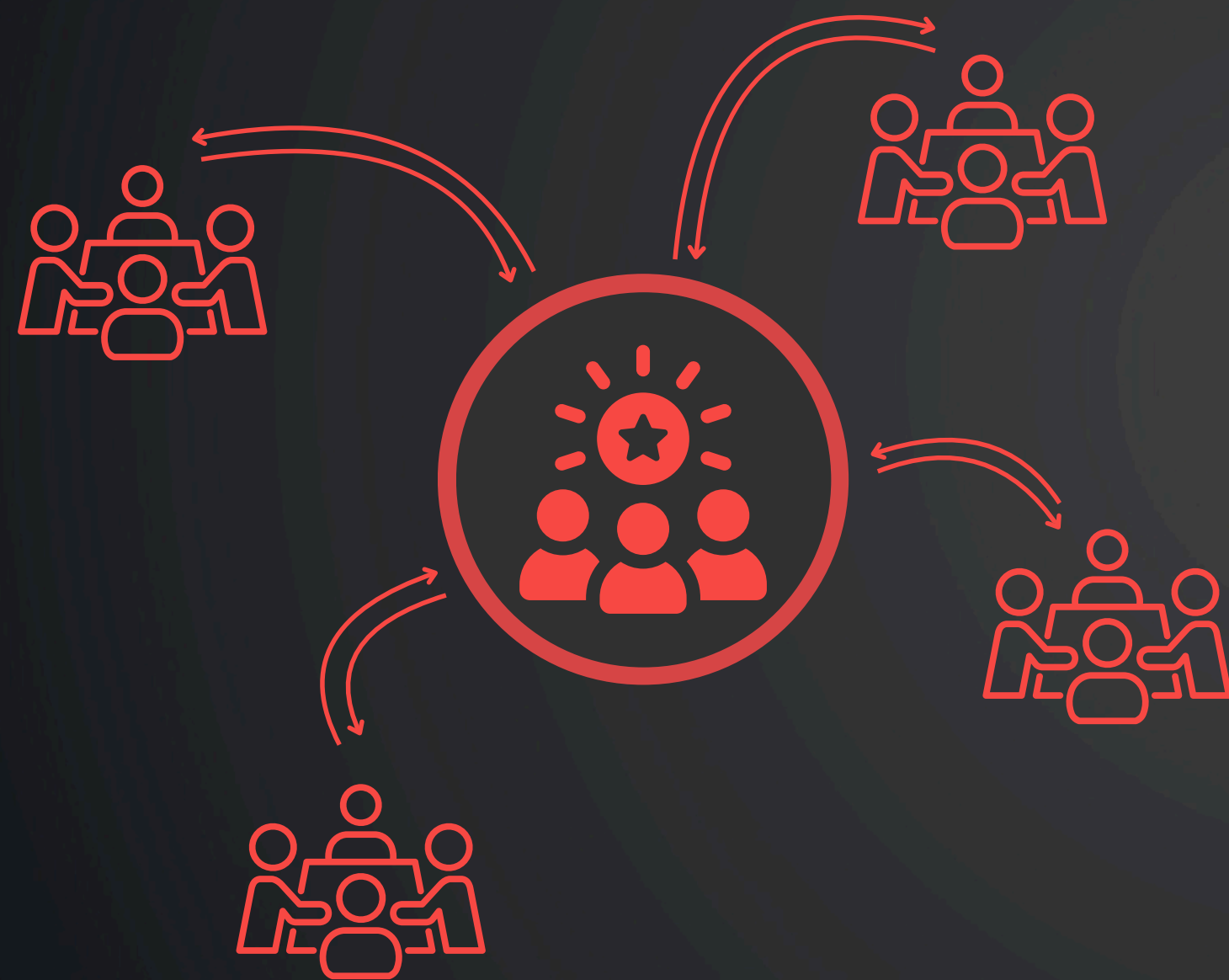
Cybersecurity center
of excellence with no
decisional power to
provide assistance to
employees in the
field.



CENTER OF EXCELLENCE^[11]

**Create a CoE with no
authority to provide
employees with best
practices for security**

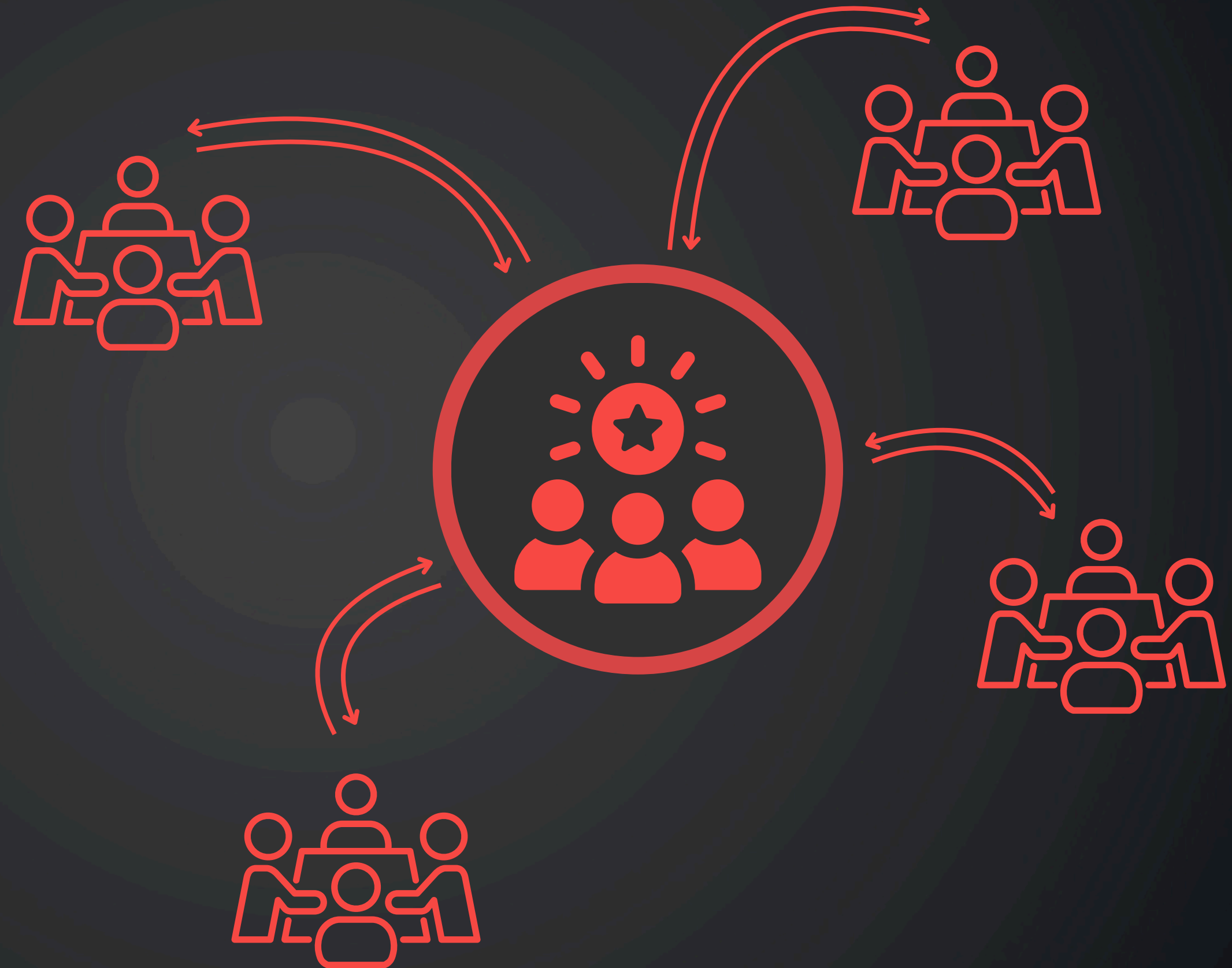
COMPARISON WITH ENISA^[13]



CENTER OF EXCELLENCE

Advisory and support entities for Cabals

- **Actionable guidelines** for teams
- Manage reports in event of a security incident
- Sharing latest innovation and technology



ABOUT RESILIENCE...

Why do we think this is the best solution?

- Shines in coping with an adverse event
- Formalization of lessons learned after a crisis (adaptation)
- Enhancement of knowledge basis (anticipation)



COE - CABAL INTERACTION

The champion initiative



THANKS FOR YOUR ATTENTION!



[4]

REFERENCES

1. [HTTPS://WWW.STATISTA.COM/STATISTICS/1480954/VALVE-EMPLOYEES/](https://www.statista.com/statistics/1480954/valve-employees/)
2. [HTTPS://X.COM/PIERSHR/STATUS/1704084070169280658](https://x.com/piershr/status/1704084070169280658)
3. MÖLLER, U., & MCCAFFREY, M. (2021). LEVELS WITHOUT BOSSES? ENTREPRENEURSHIP AND VALVE'S ORGANIZATIONAL DESIGN. THE INVISIBLE HAND IN VIRTUAL WORLDS: THE ECONOMIC ORDER OF VIDEOGAMES, ED. MATTHEW MCCAFFREY (CAMBRIDGE: CAMBRIDGE UNIVERSITY PRESS, 2021), FORTHCOMING.
4. VALVE. 2012. "HANDBOOK FOR NEW EMPLOYEES." VALVE CORPORATION, BELLEVUE, WA.
5. [HTTPS://WWW.VALVESOFTWARE.COM/EN/SECURITY](https://www.valvesoftware.com/en/security)
6. [HTTPS://STEAMDB.INFO/BLOG/VALVE-SECURITY-ONE-YEAR-ON/](https://steamdb.info/blog/valve-security-one-year-on/)
7. [HTTPS://STEAMDB.INFO/BLOG/VALVE-SECURITY-OPEN-LETTER/](https://steamdb.info/blog/valve-security-open-letter/)
8. [HTTPS://WWW.PCMAG.COM/NEWS/STEAM-STORE-SPREADED-MALWARE-AFTER-HACKER-HIJACKED-DEVELOPER-ACCOUNTS](https://www.pcmag.com/news/steam-store-spreaded-malware-after-hacker-hijacked-developer-accounts)
9. [HTTPS://BLOG.OSUM.COM/STEAM-MARKET-SHARE/](https://blog.osum.com/steam-market-share/)
10. [HTTPS://WWW.BITDEFENDER.COM/EN-US/BLOG/HOTFORSECURITY/77000-STEAM-ACCOUNTS-ARE-HACKED-AND-RAIDED-EVERY-MONTH](https://www.bitdefender.com/en-us/blog/hotforsecurity/77000-steam-accounts-are-hacked-and-raided-every-month)
11. R. L. DAFT, "ORGANIZATION THEORY & DESIGN - 13TH EDITION", CHAPTER 9 - "DESIGNS FOR DIGITAL ORGANIZATIONS AND BIG DATA ANALYTICS", 2019.
12. [HTTPS://WWW.RSISECURITY.COM/CASE-STUDY/EPIC-GAMES-CASE-STUDY/](https://www.rsisecurity.com/case-study/epic-games-case-study/)
13. [HTTPS://EUR-LEX.EUROPA.EU/ELI/REG/2019/881/OJ](https://eur-lex.europa.eu/eli/reg/2019/881/oj)